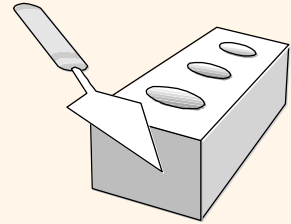


The Entity-Relationship Model

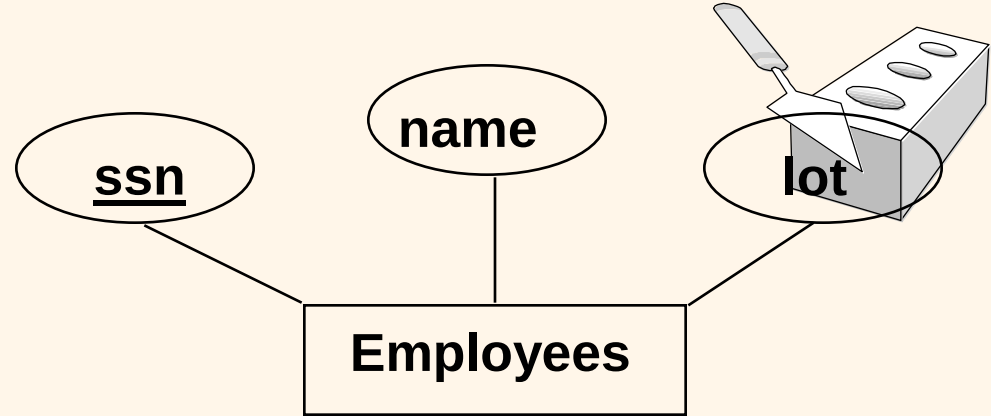
Chapter 2



Overview of Database Design

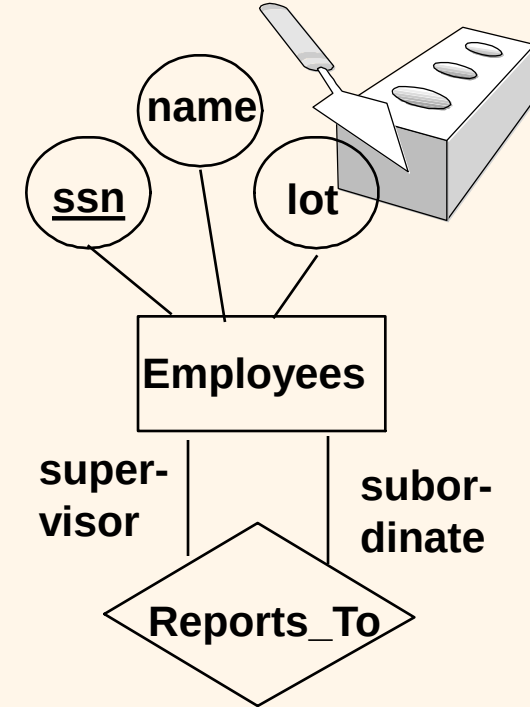
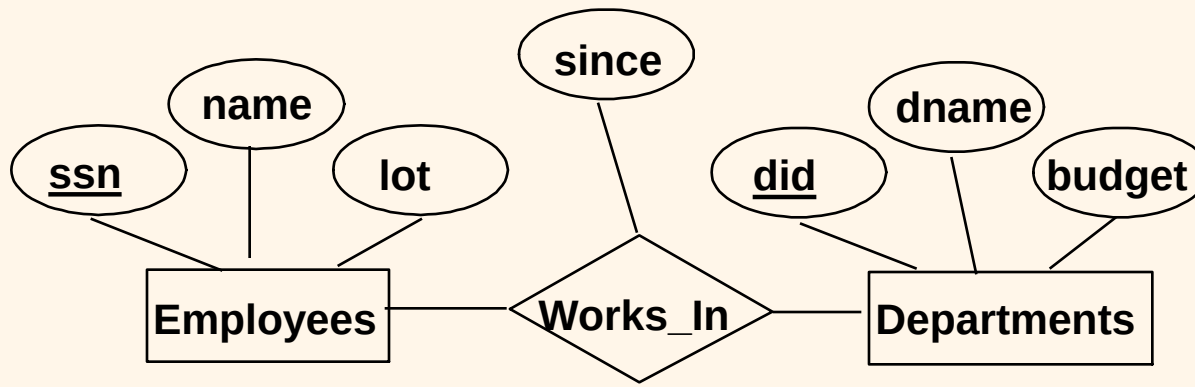
- ❖ Conceptual design: (*ER Model is used at this stage.*)
 - What are the *entities* and *relationships* in the enterprise?
 - What information about these entities and relationships should we store in the database?
 - What are the *integrity constraints* or *business rules* that hold?
 - A database 'schema' in the ER Model can be represented pictorially (*ER diagrams*).
 - Can map an ER diagram into a relational schema.

ER Model Basics



- ❖ Entity: Real-world object distinguishable from other objects. An entity is described (in DB) using a set of attributes.
- ❖ Entity Set: A collection of similar entities. E.g., all employees.
 - All entities in an entity set have the same set of attributes. (Until we consider ISA hierarchies, anyway!)
 - Each entity set has a *key*.
 - Each attribute has a *domain*.

ER Model Basics (Contd.)



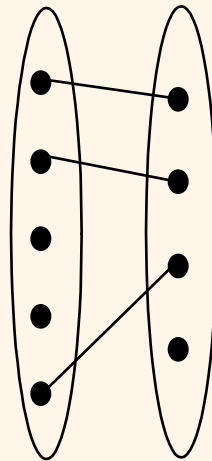
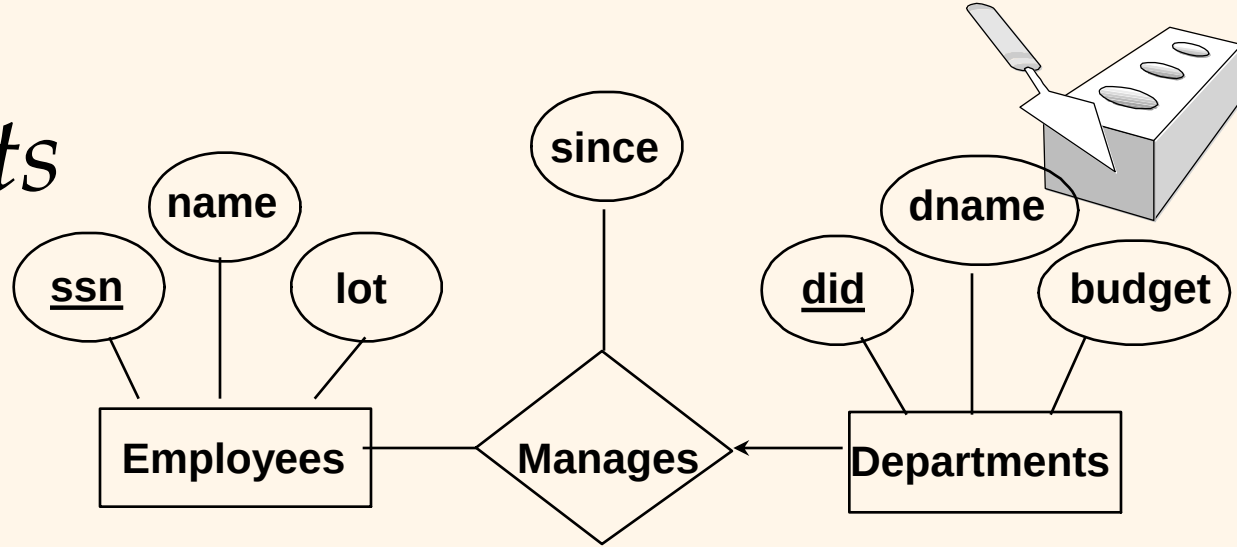
- ❖ **Relationship**: Association among two or more entities.
E.g., Attishoo works in Pharmacy department.
- ❖ **Relationship Set**: Collection of similar relationships.
 - An n-ary relationship set R relates n entity sets $E_1 \dots E_n$; each relationship in R involves entities $e_1 \in E_1, \dots, e_n \in E_n$
 - Same entity set could participate in different relationship sets, or in different “roles” in same set.

Key Constraints

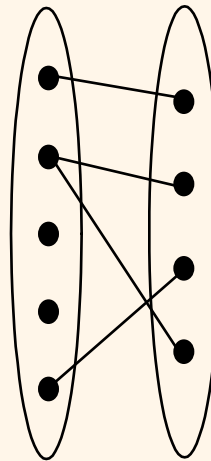
❖ Consider Works_In:

An employee can work in many departments; a dept can have many employees.

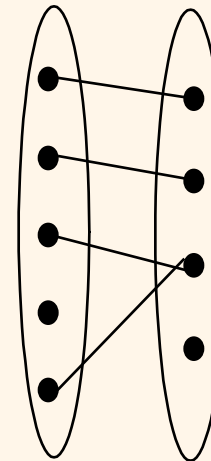
❖ In contrast, each dept has at most one manager, according to the key constraint on Manages.



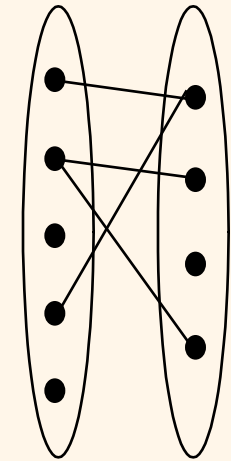
1-to-1



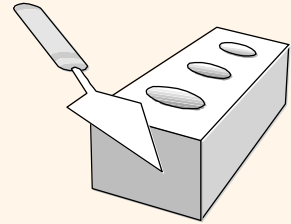
1-to Many



Many-to-1

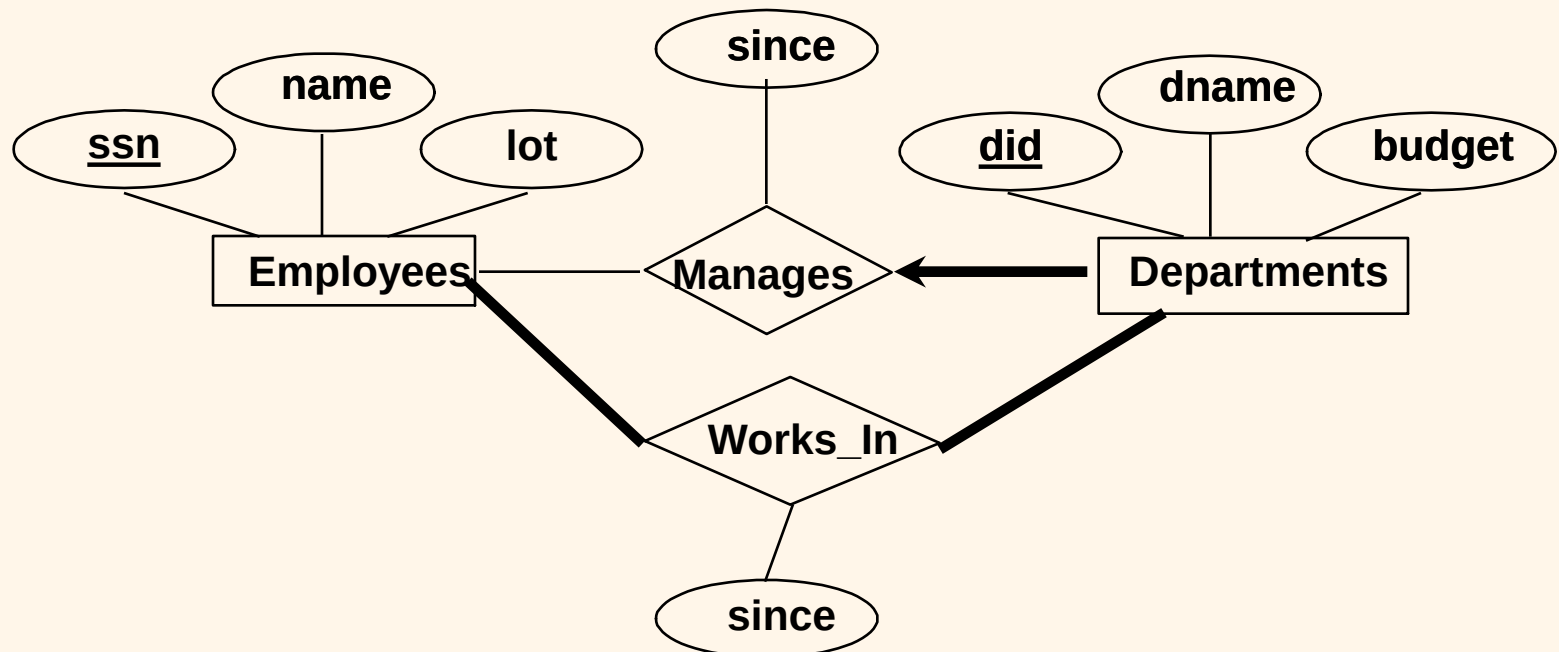


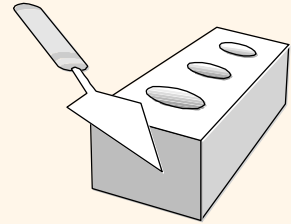
Many-to-Many



Participation Constraints

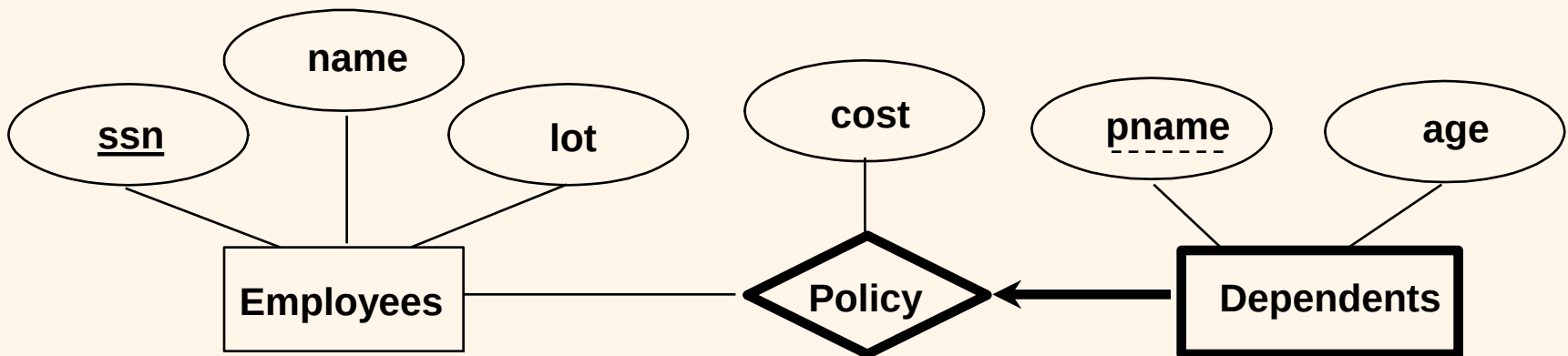
- ❖ Does every department have a manager?
 - If so, this is a participation constraint: the participation of Departments in Manages is said to be *total* (vs. *partial*).
 - Every Departments entity must appear in an instance of the Manages relationship.





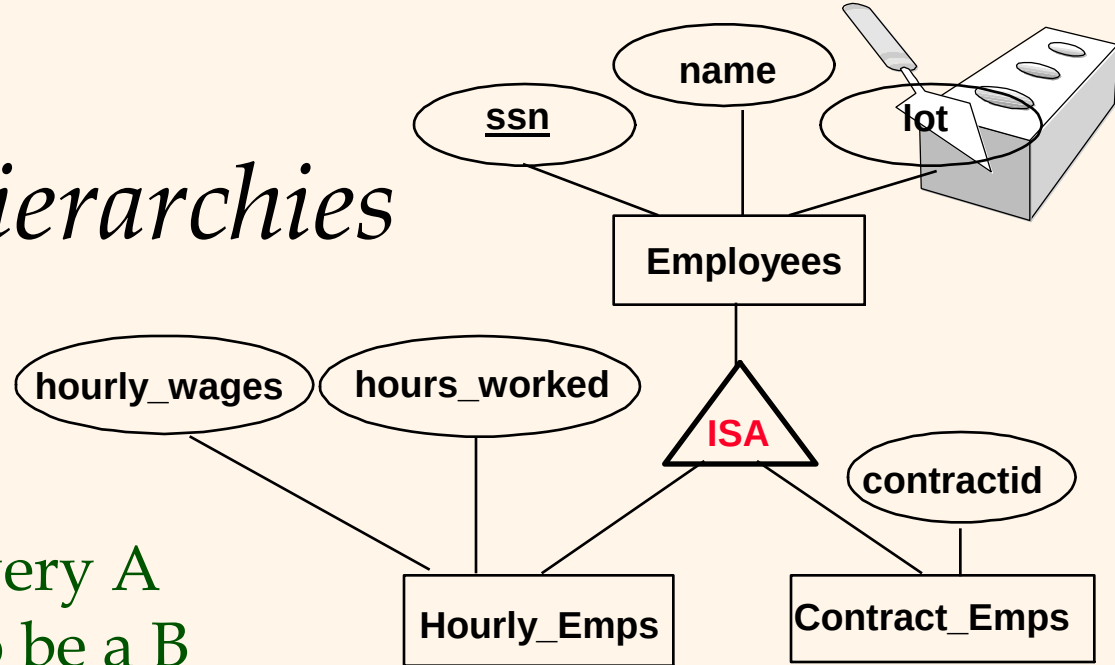
Weak Entities

- ❖ A *weak entity* can be identified uniquely only by considering the primary key of another (*owner*) entity.
 - Owner entity set and weak entity set must participate in a one-to-many relationship set (one owner, many weak entities).
 - Weak entity set must have total participation in this *identifying* relationship set.



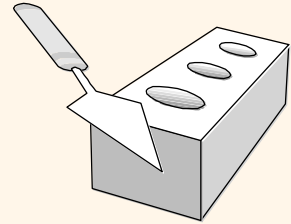
ISA ('is a') Hierarchies

- ❖ As in C++, or other PLs, attributes are inherited.
- ❖ If we declare A **ISA** B, every A entity is also considered to be a B entity.



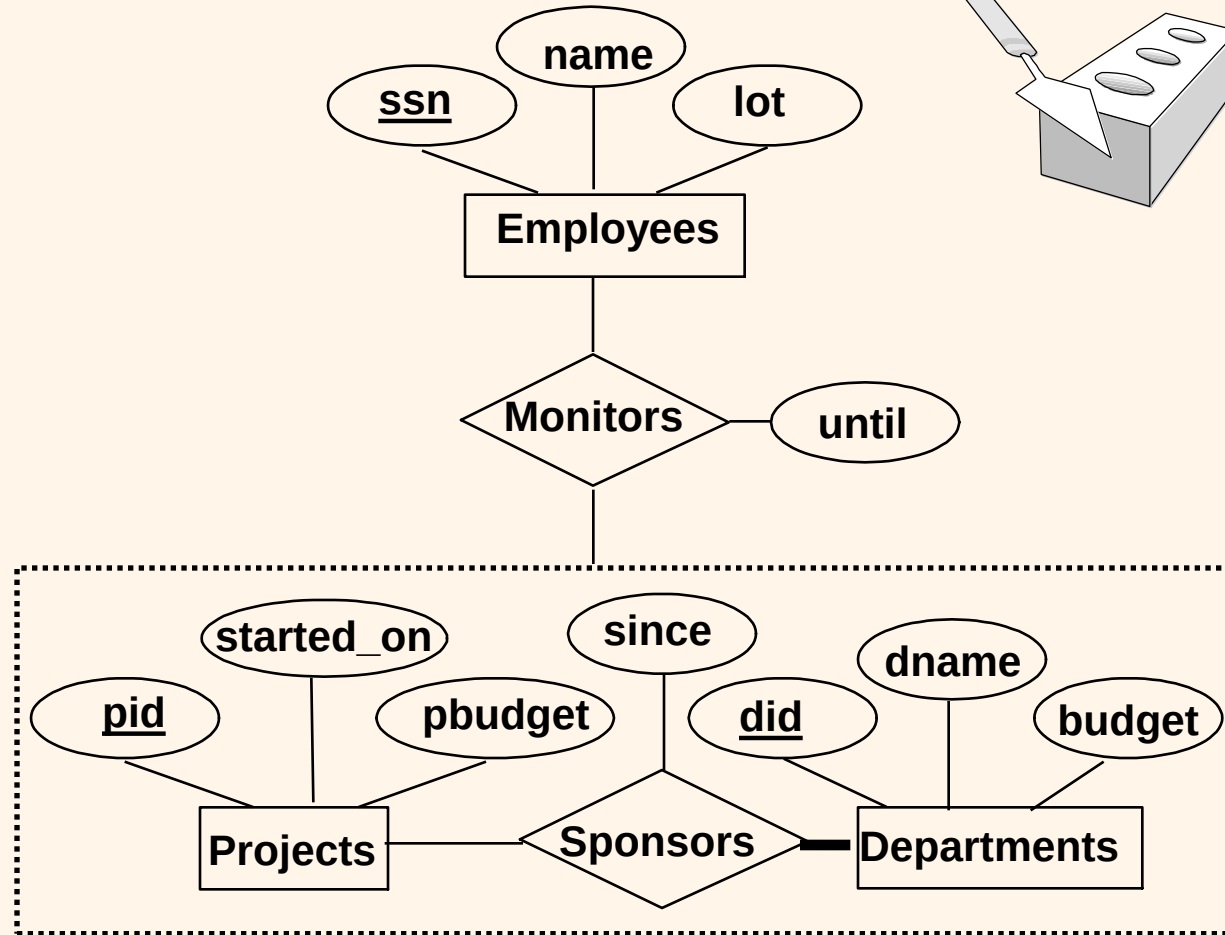
- ❖ *Overlap constraints*: Can Joe be an Hourly_Emps as well as a Contract_Emps entity? (*Allowed/disallowed*)
- ❖ *Covering constraints*: Does every Employees entity also have to be an Hourly_Emps or a Contract_Emps entity? (*Yes/no*)
- ❖ Reasons for using ISA:
 - To add descriptive attributes specific to a subclass.
 - To identify entities that participate in a relationship.

Aggregation

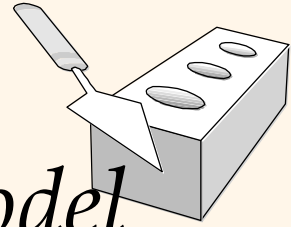


- ❖ Used when we have to model a relationship involving (entity sets and) a *relationship set*.

- Aggregation allows us to treat a relationship set as an entity set for purposes of participation in (other) relationships.



- * *Aggregation vs. ternary relationship:*
 - ❖ Monitors is a distinct relationship, with a descriptive attribute.
 - ❖ Also, can say that each sponsorship is monitored by at most one employee.



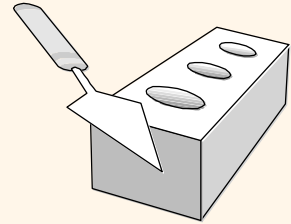
Conceptual Design Using the ER Model

❖ Design choices:

- Should a concept be modeled as an entity or an attribute?
- Should a concept be modeled as an entity or a relationship?
- Identifying relationships: Binary or ternary? Aggregation?

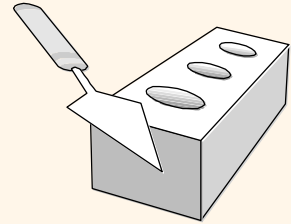
❖ Constraints in the ER Model:

- A lot of data semantics can (and should) be captured.
- But some constraints cannot be captured in ER diagrams.



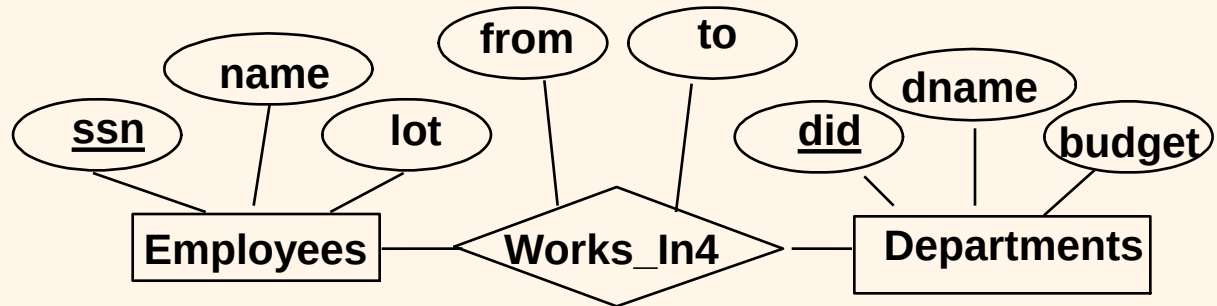
Entity vs. Attribute

- ❖ Should *address* be an attribute of Employees or an entity (connected to Employees by a relationship)?
- ❖ Depends upon the use we want to make of address information, and the semantics of the data:
 - If we have several addresses per employee, *address* must be an entity (since attributes cannot be set-valued).
 - If the structure (city, street, etc.) is important, e.g., we want to retrieve employees in a given city, *address* must be modeled as an entity (since attribute values are atomic).

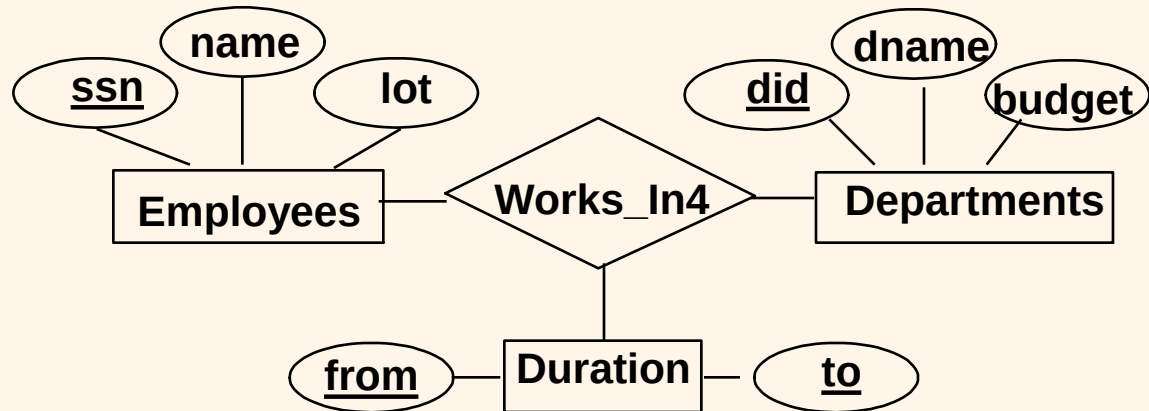


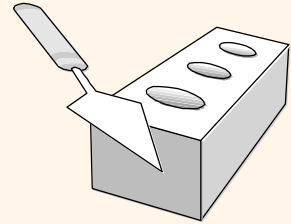
Entity vs. Attribute (Contd.)

- ❖ Works_In4 does not allow an employee to work in a department for two or more periods.



- ❖ Similar to the problem of wanting to record several addresses for an employee: We want to record *several values of the descriptive attributes for each instance of this relationship*. Accomplished by introducing new entity set, **Duration**.

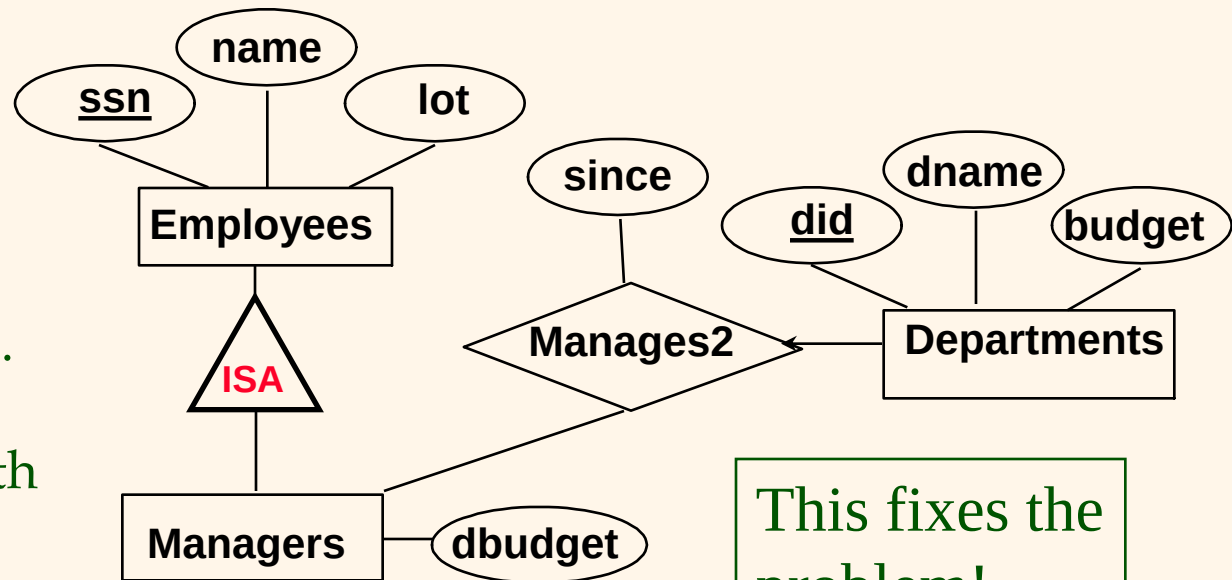
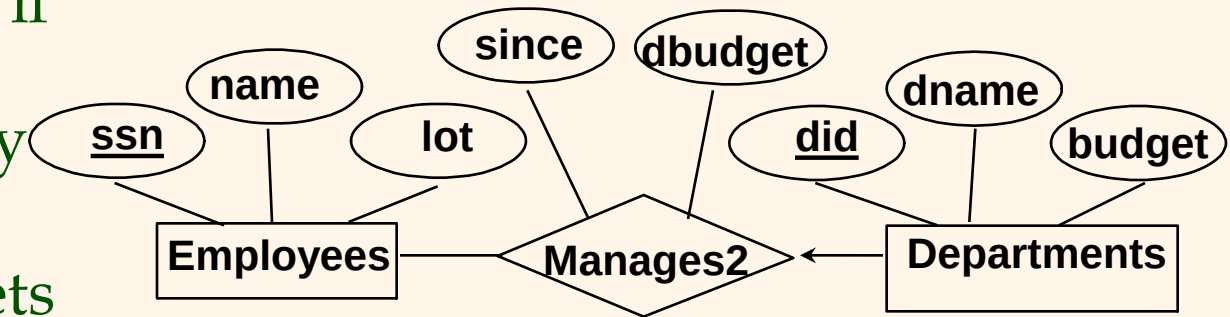




Entity vs. Relationship

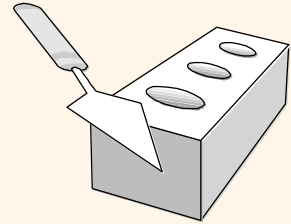
- ❖ First ER diagram OK if a manager gets a separate discretionary budget for each dept.
- ❖ What if a manager gets a discretionary budget that covers *all* managed depts?

- **Redundancy:** *dbudget* stored for each dept managed by manager.
- **Misleading:** Suggests *dbudget* associated with department-mgr combination.

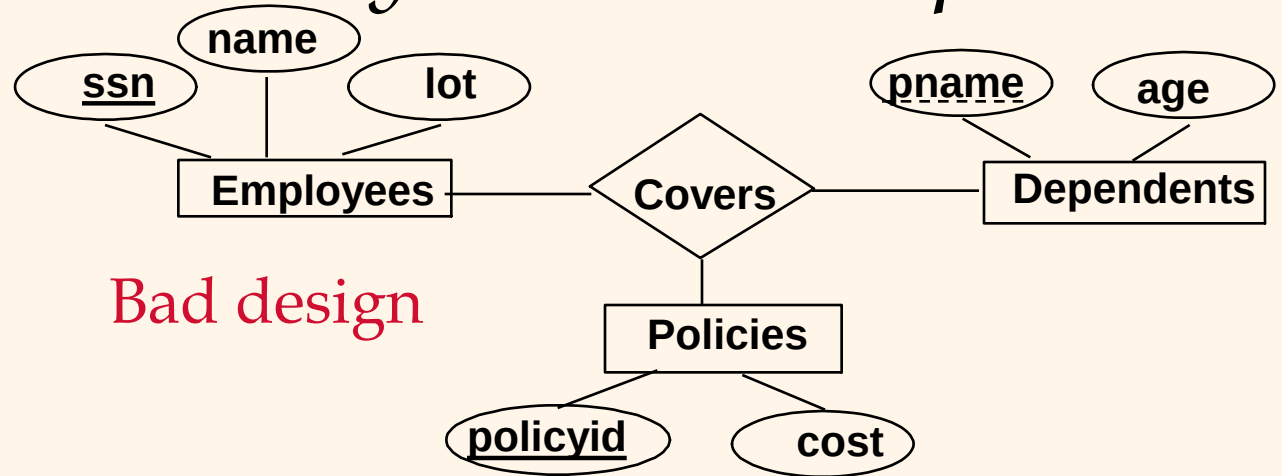


This fixes the problem!

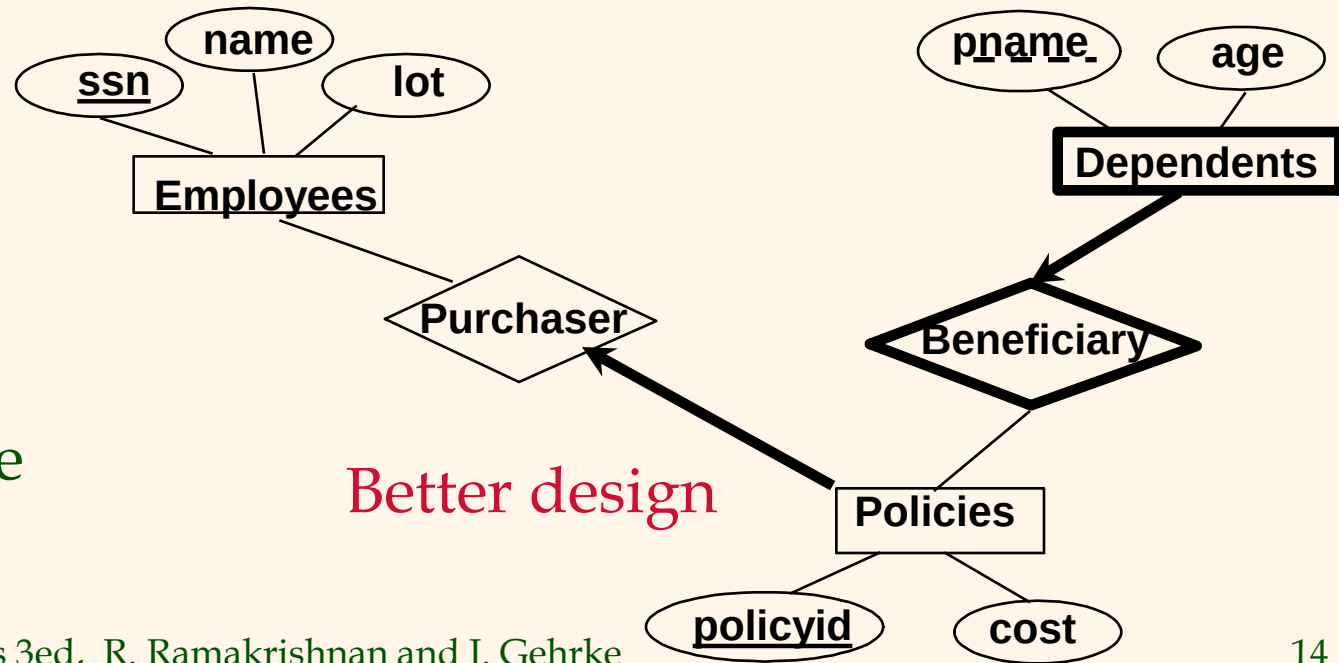
Binary vs. Ternary Relationships

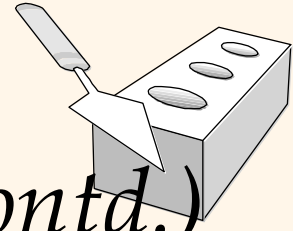


- ❖ If each policy is owned by just 1 employee, and each dependent is tied to the covering policy, first diagram is inaccurate.



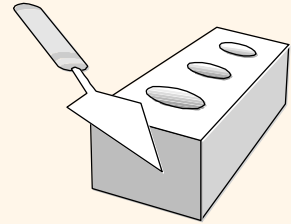
- ❖ What are the additional constraints in the 2nd diagram?





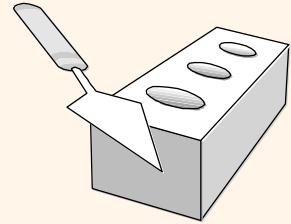
Binary vs. Ternary Relationships (Contd.)

- ❖ Previous example illustrated a case when two binary relationships were better than one ternary relationship.
- ❖ An example in the other direction: a ternary relation **Contracts** relates entity sets **Parts**, **Departments** and **Suppliers**, and has descriptive attribute *qty*. No combination of binary relationships is an adequate substitute:
 - S “can-supply” P, D “needs” P, and D “deals-with” S does not imply that D has agreed to buy P from S.
 - How do we record *qty*?



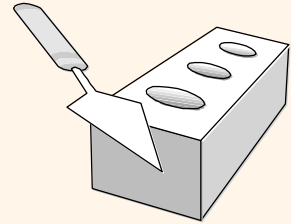
Summary of Conceptual Design

- ❖ *Conceptual design follows requirements analysis,*
 - Yields a high-level description of data to be stored
- ❖ ER model popular for conceptual design
 - Constructs are expressive, close to the way people think about their applications.
- ❖ Basic constructs: *entities, relationships, and attributes* (of entities and relationships).
- ❖ Some additional constructs: *weak entities, ISA hierarchies, and aggregation.*
- ❖ Note: There are many variations on ER model.



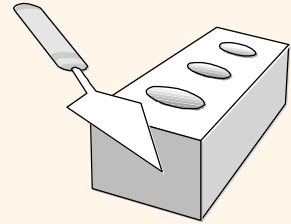
Summary of ER (Contd.)

- ❖ Several kinds of integrity constraints can be expressed in the ER model: *key constraints, participation constraints, and overlap/covering constraints* for ISA hierarchies. Some *foreign key constraints* are also implicit in the definition of a relationship set.
 - Some constraints (notably, *functional dependencies*) cannot be expressed in the ER model.
 - Constraints play an important role in determining the best database design for an enterprise.



Summary of ER (Contd.)

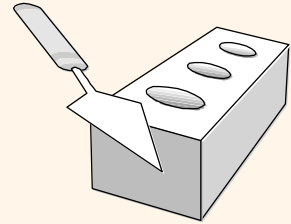
- ❖ ER design is *subjective*. There are often many ways to model a given scenario! Analyzing alternatives can be tricky, especially for a large enterprise. Common choices include:
 - Entity vs. attribute, entity vs. relationship, binary or n-ary relationship, whether or not to use ISA hierarchies, and whether or not to use aggregation.
- ❖ Ensuring good database design: resulting relational schema should be analyzed and refined further. FD information and normalization techniques are especially useful.



Example

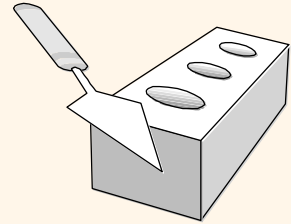
- ❖ A University database contains information about **professors** (identified by social security number, or *SSN*) and **courses** (identified by *courseid*). Professors teach courses.
- ❖ Each of the following situations concerns the **Teaches** relationship set. For each situation, draw an ER diagram that describes it (assuming that no further constraints hold).

Example (cont')

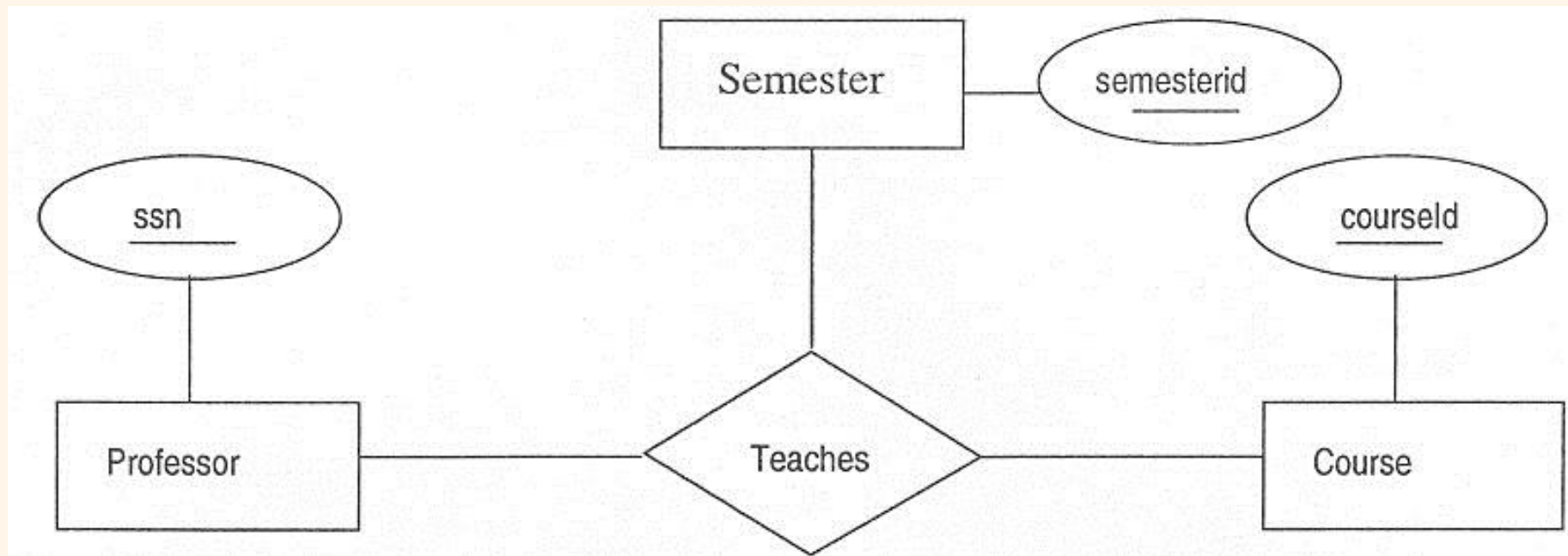


- 1) Professors can teach the same course in several semesters, and each offering must be recorded.
- 2) Professors can teach the same course in several semesters, and only the most recent such offering needs to be recorded. (Assume this condition applies to all subsequent questions.)
- 3) Every professor must teach some course
- 4) Every professor teaches exactly one course (no more, no less).
- 5) Every professor teaches exactly one course (no more no less), and every course must be taught by some professor.
- 6) Now assume that certain courses can be taught by a team of professors jointly, but it is possible that no one professor in a team can teach the course. Model this situation introducing additional entity sets and relationship sets if necessary.

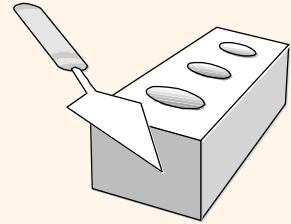
Example (cont')



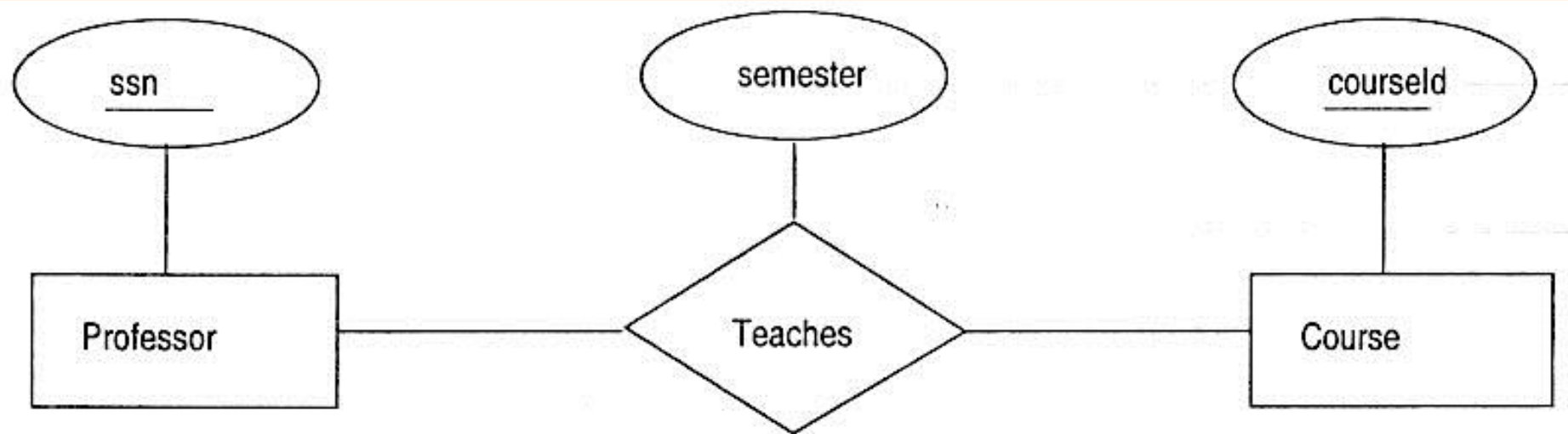
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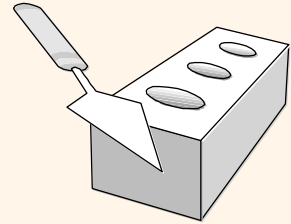
Example (cont')



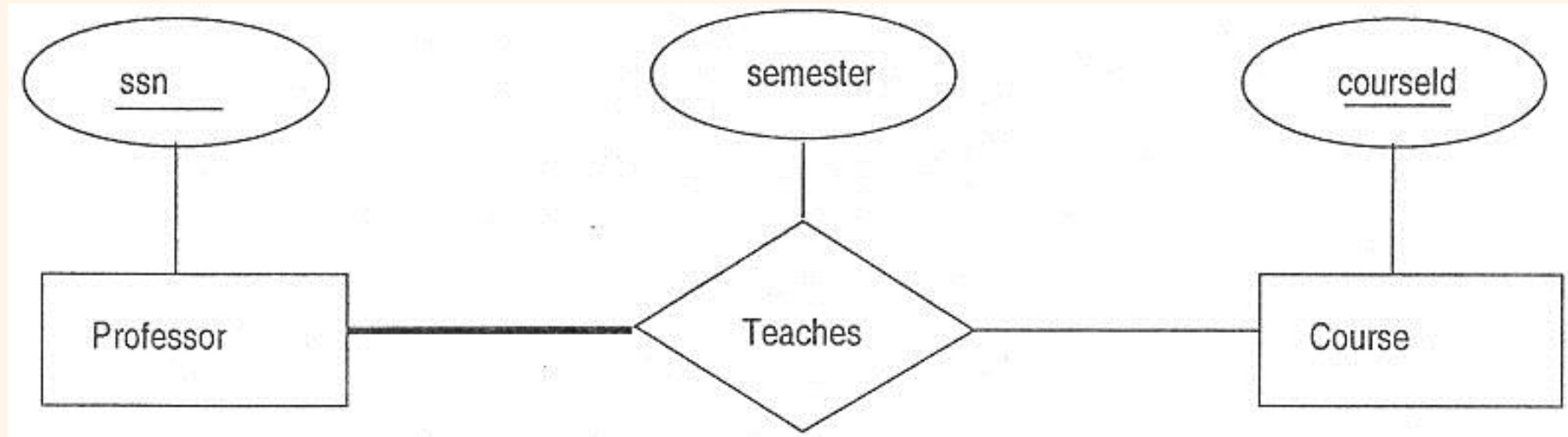
- 2) Professors can teach the same course in several semesters, and only the most recent such offering needs to be recorded. (Assume this condition applies to all subsequent questions.)



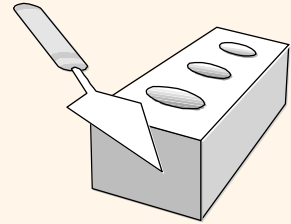
Example (cont')



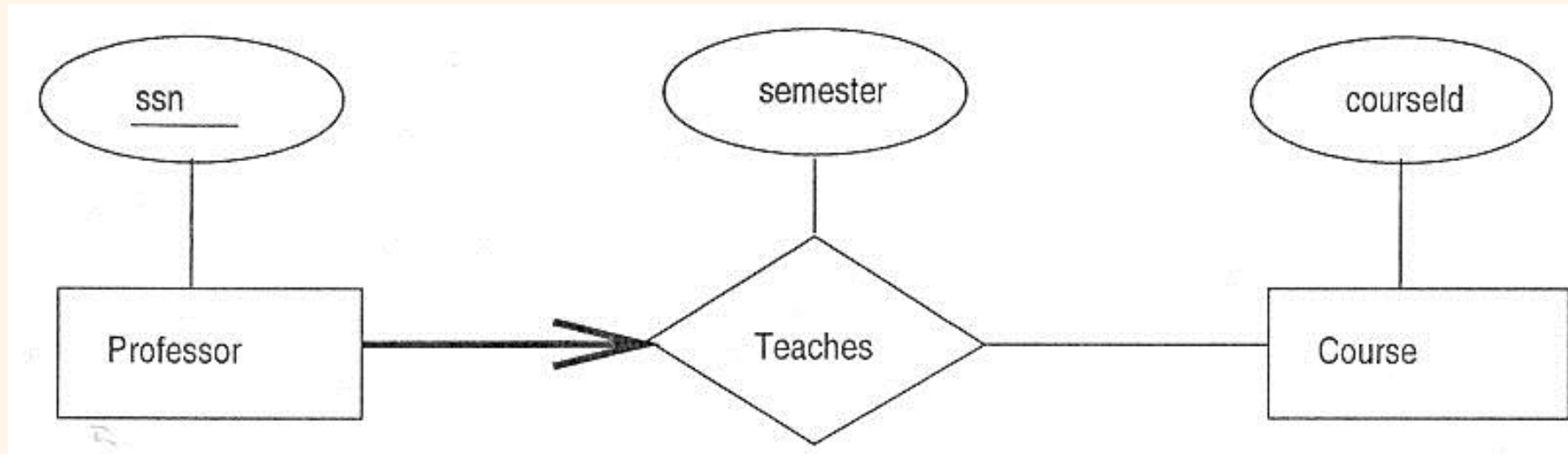
3) Every professor must teach some course



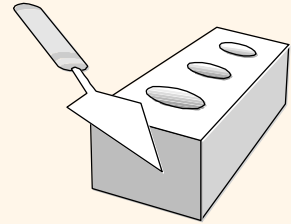
Example (cont')



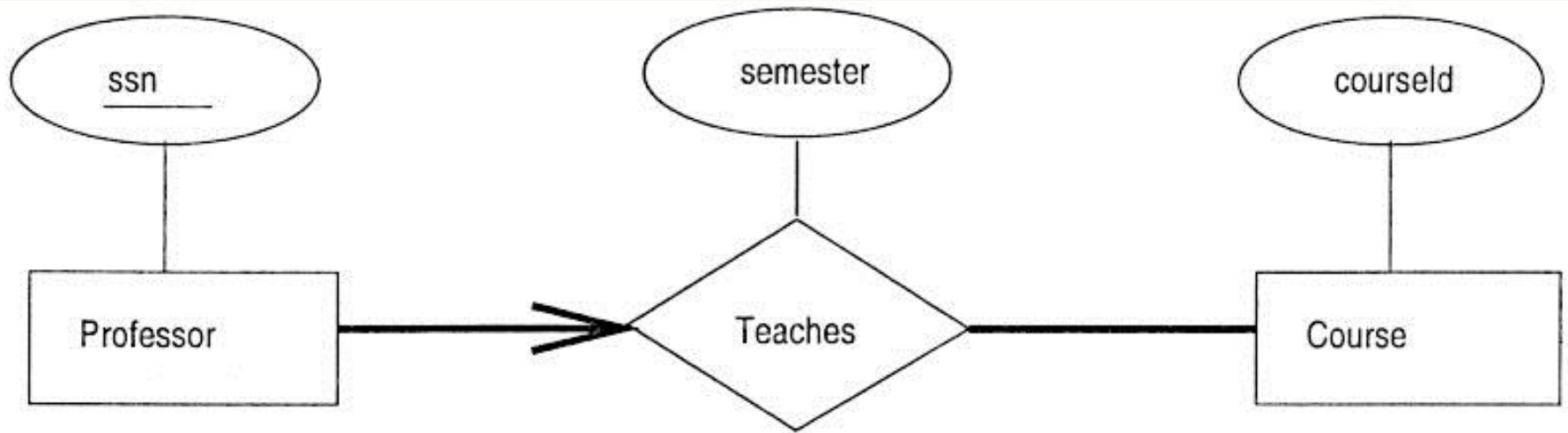
- 4) Every professor teaches exactly one course (no more, no less).



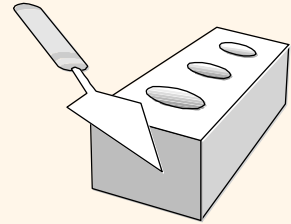
Example (cont')



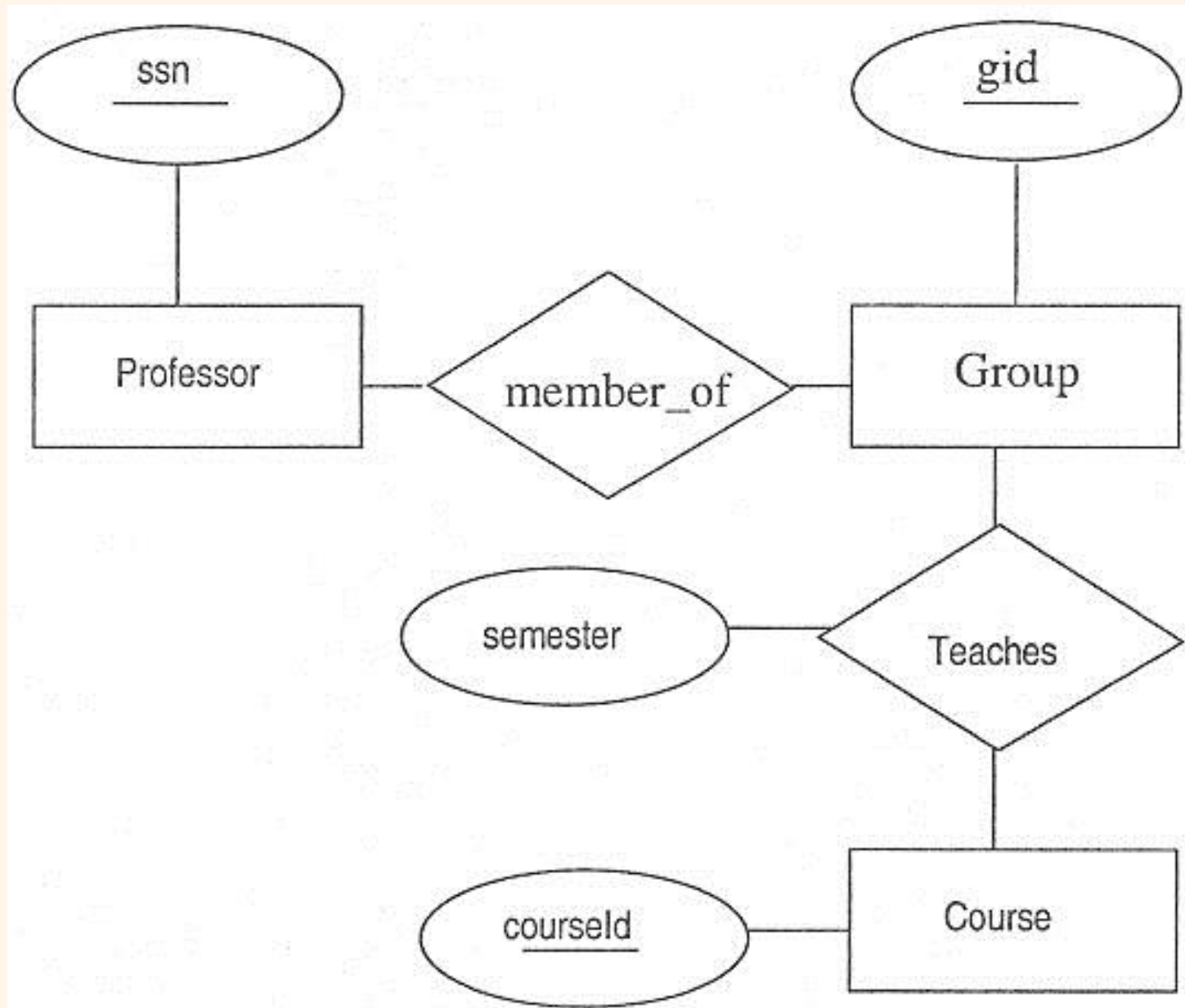
- 5) Every professor teaches exactly one course (no more no less), and every course must be taught by some professor.



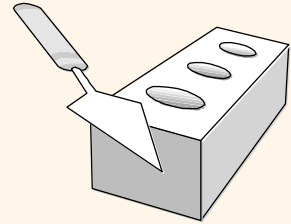
Example (cont')



- 6) Now assume that certain courses can be taught by a team of professors jointly, but it is possible that no one professor in a team can teach the course. Model this situation introducing additional entity sets and relationship sets if necessary.



Example



- ❖ A company database needs to store information about **employees** (identified by *ssn*, with *salary* and *phone* as attributes); **departments** (identified by *dno*, with *dname* and *budget* as attributes); and **children** of employees (with *name* and *age* as attributes). Employees work in departments; each department is managed by an employee; a child must be identified uniquely by name when the parent (who is an employee; assume that only one parent works for the company) is known. We are not interested in information about a child once the parent leaves the company.

